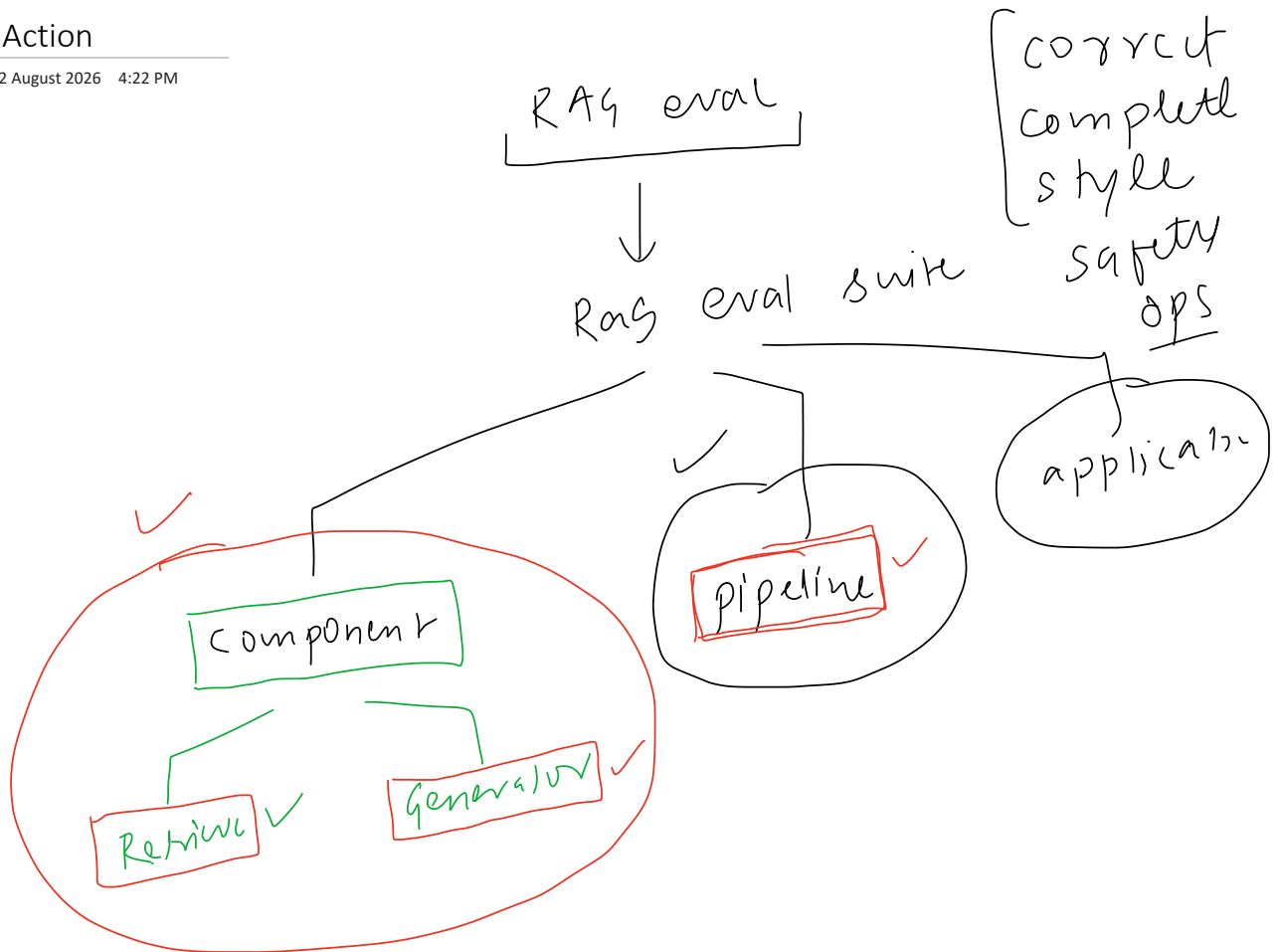


Plan of Action

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Creating the Generator

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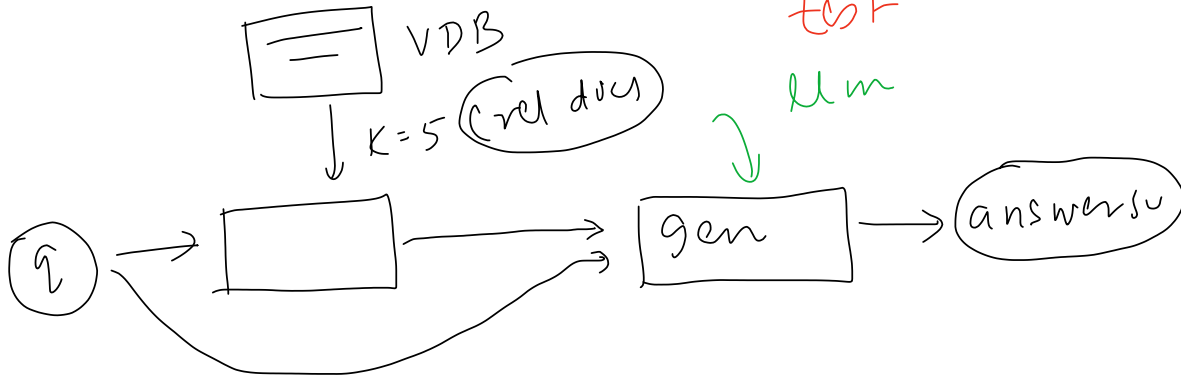
[Generator]

rag app

compm



test
llm



Generator Failure Modes

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unfaithful response

Query

Does the CampusX AI Engineering program include live classes?



faithfulness

→ co

→ corr

Retrieved context

The AI Engineering program includes recorded lessons, coding assignments, projects, and weekly doubt-solving sessions.

→ context

Yes. The program includes two live classes every week along with weekly doubt-solving sessions.

→ answer



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2026 4:23 PM

[dataset]

generator

Query: what does it mean for a benchmark to get saturated?

Saturation = scores cluster high & close together (95%, 94%, 92%)
→ can't tell models apart → benchmark no longer useful

"A benchmark gets saturated when it can no longer distinguish between models. This happens once models all score very high on it, and the models show a wide range of scores across the benchmark."

decompose into clair

Claim 1 — a saturated benchmark no longer distinguishes models
context says exactly this ("can't differentiate the models")

Claim 2 — it happens once models all score very high
context supports this (95% / 94% / 92% clustering)

Claim 3 — models show a wide range of scores
 CONTRADICTS context — saturation is clustering, not spreading

supported $\div$ total = $2 \div 3$

Faithfulness = 0.67

powerful

judge

40 /

Query: what does it mean for a benchmark to get saturated?

relevance checks every statement against THIS — not against the context

"A benchmark gets saturated when model scores cluster high and close together, so it can no longer tell models apart. Separately, benchmark contamination happens when test data leaks into training data."

split into statements

Statement 1 — saturation is when scores cluster high & close
directly answers the query

Statement 2 — so the benchmark can't tell models apart
directly answers the query

Statement 3 — benchmark contamination is test data leaking
OFF-TOPIC — true & in the context, but doesn't answer "saturation"

$$\text{relevant} \div \text{total} = 2 \div 3$$

Answer relevance = 0.67

question

golden context

what is rag triad

what are online evals

LLM

3/5

A hand-drawn diagram illustrating a relationship. On the left, a rectangular box contains the word "Zehri". An arrow points upwards towards this box. A horizontal line connects the "Zehri" box to a larger rectangular box on the right. This larger box contains the word "gen". The "gen" box is enclosed within a hand-drawn circle.

$q / \text{context} \rightarrow \text{generator}$
 $\downarrow$
golden
answer

A diagram illustrating the process of generating an answer from a question. On the left, a green circle contains the letter 'a', with two green arrows pointing towards it from the left. Below this circle, the word 'answe' is written in green and underlined. To the right of the circle, three green lines radiate outwards towards a green rectangular box labeled 'LLM'. From the bottom of the 'LLM' box, a green arrow points down towards the word 'answe'.

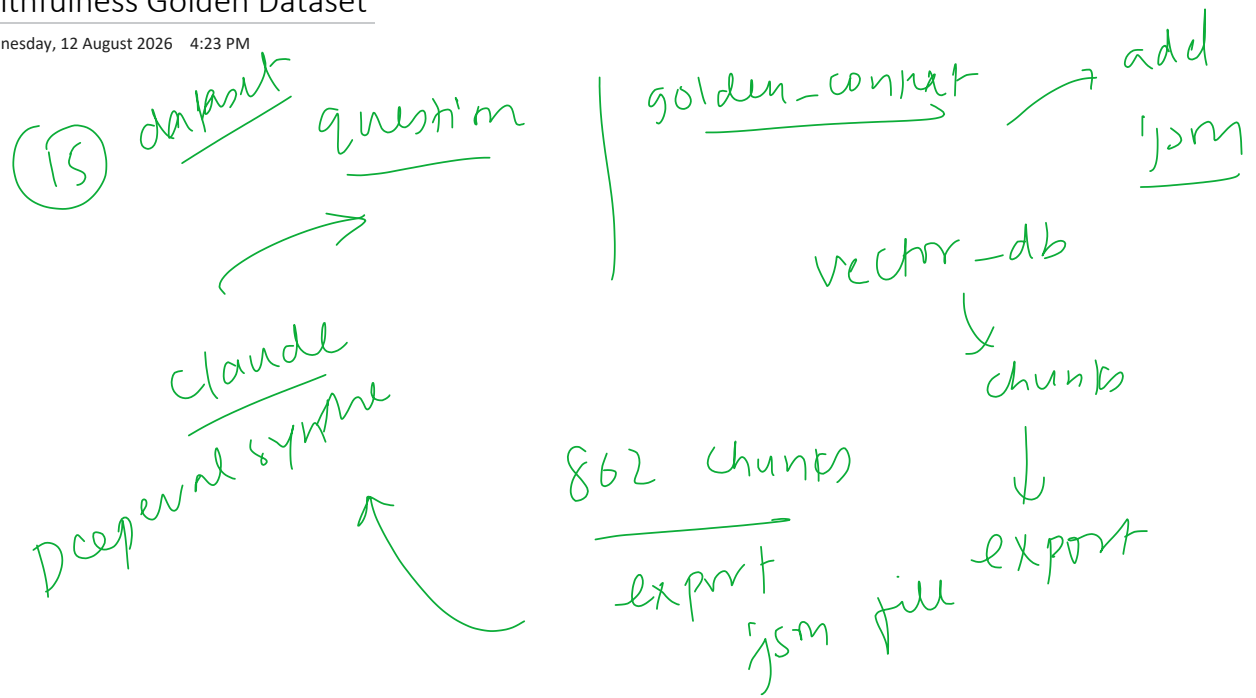
→ claims

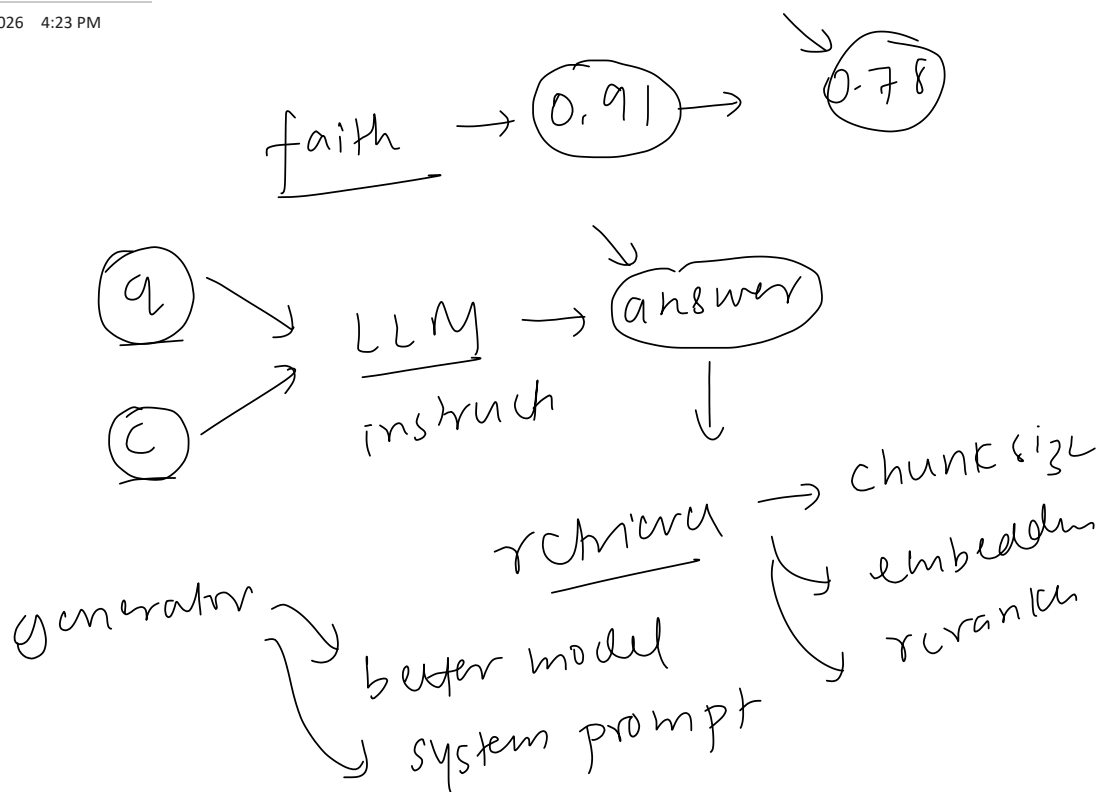
- C1 ✓
- C2 ✓
- C5 ✗

$$\frac{2}{3}$$

Faithfulness Golden Dataset

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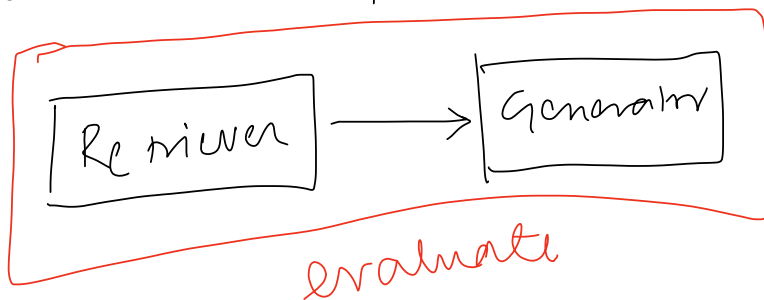




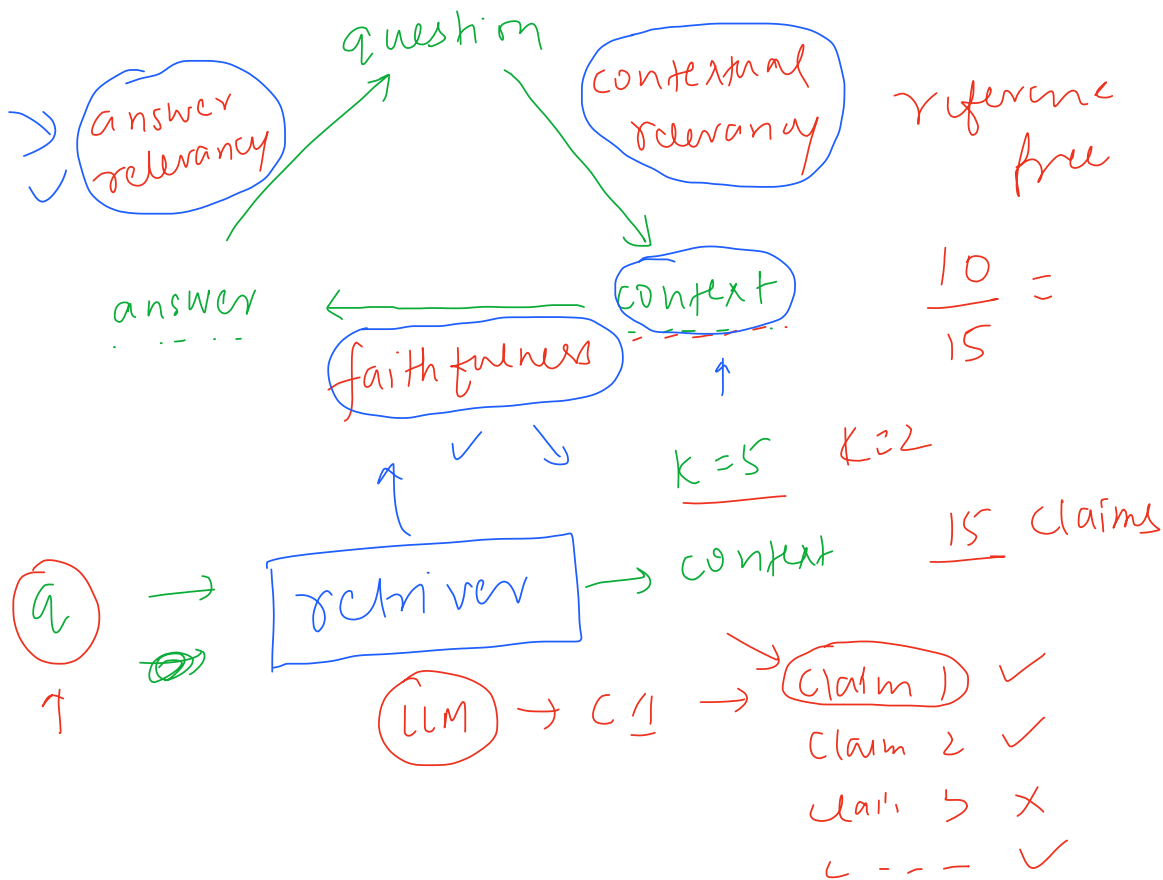
Building the RAG Pipeline

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RAG pipeline



- ✓ step 1 — build this pipeline
- step 2 — evaluate



Query (g006): if a model tops the benchmarks, why not just pick it?
needs the Zomato model-A-vs-B custom-eval comparison

Retrieved context (this is what we score — the 5 fetched chunks)

1. "model A is top of the table, so the obvious pick seems model A..."
2. "...run both on your own data, accuracy of A is 94%, B is 91%..."
3. "benchmark contamination is when test data leaks into training..."
4. "you go to the leaderboard, filter down to your top candidates..."
5. "benchmarks measure a model's knowledge and reasoning ability..."

judge each retrieved chunk against the query

Chunk 1 — model A tops the table, seems the obvious pick

✓

Chunk 2 — on your own data the accuracy gap is small

✓

Chunk 3 — benchmark contamination (off-topic)

✗

Chunk 4 — how to read an LLM leaderboard (off-topic)

✗

Chunk 5 — what benchmarks measure (off-topic)

✗



relevant ÷ total = 2 ÷ 5
Context relevance = 0.40

faith $\rightarrow$ 0.93
ans rel $\rightarrow$ 0.86
[contextual
relevance $\rightarrow$ 0.42]

retriever

+ recall - 99
+ precision - 87 $\frac{42\%}{2/5}$

$K=5$

✓ C1 $\rightarrow$ 5 lines $\rightarrow$ 5 claims
✓ C2 $\rightarrow$ 5 cla. 1-4

✓ C3

✓ C4

✗ C5